

HR Analytics Dashboard

Power BI Project Report

A comprehensive HR analytics solution built with Power BI, Power Query, and DAX to analyze workforce data across 1,000 employee records.

Project	HR Analytics Dashboard
Tool	Microsoft Power BI Desktop
Dataset	employees_data.csv (1,000 records)
Report Pages	3 (Overview, Attrition, Salary & Performance)
Date	May, 2026

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1. Executive Summary

This report presents a full documentation of the HR Analytics Dashboard, a multi-page interactive report developed in Microsoft Power BI. The project was completed as both an academic and professional portfolio deliverable, with the aim of demonstrating competency in data cleaning, DAX measure development, data visualization, and analytical storytelling.

The dashboard was built using a synthetic HR dataset containing 1,000 employee records across five departments. The final report consists of three interconnected pages: an HR Analytics Overview page covering workforce demographics and hiring trends; an Attrition Analysis page examining employee turnover patterns; and a Salary & Performance page exploring compensation and performance relationships.

Key findings from the analysis include a notably high overall attrition rate of 50.30%, a near-equal gender split, a workforce skewed toward the older age group (45+), and a weak relationship between performance scores and salary levels. These findings, while based on a limited and likely synthetic dataset, provide a foundation for exploring workforce management strategies and may indicate areas warranting further investigation in a real organizational context.

Metric	Value
Total Employees	1,000
Average Salary	\$55,369
Attrition Rate	50.30%
Average Experience Years	11.46
High Performance Employees	380 (38.0%)
Largest Department	Marketing (211 employees, 21.1%)
Highest Attrition Department	Finance (54.00%)
Gender Distribution	Female 50.9% / Male 49.1%

2. Introduction

Human Resources analytics has become a critical function in modern organizations. By applying data analysis techniques to workforce data, HR teams can make more informed decisions about recruitment, retention, compensation, and employee development. Power BI is one of the leading business intelligence tools used to build interactive, real-time dashboards that support such decision-making.

This project was undertaken to apply and demonstrate a range of data analytics skills within a realistic HR context. The objective was to take a raw HR dataset, clean and transform it, develop relevant metrics using DAX, and present findings through a well-structured, multi-page Power BI report.

The report was originally designed as a single-page dashboard. However, after an initial review, it became clear that a multi-page structure would better serve both readability and analytical depth. The final design separates the content into three focused pages, allowing the reader to explore workforce overview, attrition patterns, and salary and performance independently, while maintaining cross-page consistency through shared slicers and a unified visual theme.

Project Objectives

- Analyze the overall workforce composition across departments, gender, and age groups.
- Identify attrition patterns and risk factors by salary level, gender, department, and experience.
- Examine compensation levels across job titles and departments.
- Investigate the relationship between salary and performance score.
- Present findings through a professional, interactive Power BI report.

Skills Demonstrated

This project demonstrates the following technical competencies:

- Power BI: Multi-page dashboard design, interactive slicers, navigation buttons, scatter plots, box plots, KPI cards, and visual formatting.
- Power Query: Data type conversion, duplicate removal, column creation using conditional logic and date functions.
- DAX: Calculated measures including attrition rate, average salary, percentage calculations, and filtered aggregations.
- Data Modeling: Structuring an analytical model to support multiple related measures and visualizations.
- Analytical Storytelling: Organizing insights into a coherent narrative across three focused report pages.

3. Dataset Description

The dataset used in this project is a synthetic HR dataset named `employees_data.csv`, containing 1,000 employee records. It was structured to reflect typical HR data used in workforce analytics, including demographic information, employment details, compensation, performance, and attrition status.

Column Name	Data Type	Description
EmployeeID	Integer	Unique identifier for each employee
Name	Text	Employee full name
Gender	Text	Employee gender (Male / Female)
Age	Integer	Employee age in years
Department	Text	Department the employee belongs to
JobTitle	Text	Employee job role/title
HireDate	Date	Date the employee was hired
Salary	Integer	Annual salary in USD
PerformanceScore	Integer	Employee performance rating (1–5 scale)
Attrition	Text	Whether the employee has left (Yes / No)

The dataset covers five departments: Marketing, HR, Finance, Sales, and IT. Salary values range from below \$40,000 to above \$70,000, and performance scores are recorded on a 1 to 5 scale. The Attrition column indicates whether an employee has left the organization, making it the primary variable of interest for the attrition analysis page.

It should be noted that the dataset appears to be synthetically generated for educational and demonstration purposes. Some observed patterns, such as the unusually high attrition rate of 50.30%, may not be representative of real-world HR data and should be interpreted with this context in mind.

4. Data Preparation

Before building the report, the dataset was cleaned and transformed using Power Query Editor in Power BI Desktop. The following steps were completed to ensure data quality and to create the derived columns required for analysis.

4.1 Data Cleaning

Remove Duplicates

Duplicate records were removed using the EmployeeID column as the unique key. This step ensures that each employee appears only once in the dataset and that all aggregated metrics are accurate.

Date Format Conversion

The HireDate column was converted from text format to the Date data type. This step was essential for calculating employee tenure and for the Employee Hiring by Year trend chart.

4.2 Derived Columns

ExperienceYears

A new column called ExperienceYears was created to calculate the number of years each employee has been with the company. The calculation is based on the difference between the current date and the HireDate column, converted to a whole number of years. This column is used in the attrition by experience analysis.

SalaryLevel

A categorical column called SalaryLevel was created using conditional logic to group employees by compensation level:

- Low: Annual salary below \$40,000
- Medium: Annual salary between \$40,000 and \$70,000
- High: Annual salary above \$70,000

This classification allows for comparison of attrition rates across compensation groups and provides a higher-level view of salary distribution.

AgeGroup

A categorical column called AgeGroup was created to segment employees by age:

- Young: Employees aged 18 to 29
- Middle Age: Employees aged 30 to 44

- Older: Employees aged 45 and above

This segmentation is used in the age distribution chart and in the experience distribution by age group box plot.

5. DAX Measures

The following DAX measures were created in Power BI to support the dashboard calculations, KPI cards, and comparative analyses across all three report pages.

Measure Name	Description	Primary Usage
Total Employees	Count of all employee records	KPI Card, context filters
Average Salary	Mean salary across all employees	KPI Card, department comparison
Total Attrition	Count of employees where Attrition = Yes	Attrition Analysis page
Attrition Rate	Total Attrition / Total Employees	KPI Card, department/gender breakdown
Average Experience Years	Mean of ExperienceYears column	KPI Card, experience analysis
High Performance Employees	Count of employees with PerformanceScore >= 4	KPI Card
Average Salary by Performance Score	Average salary grouped by performance	Salary vs Performance scatter
Female Percentage	Percentage of employees who are Female	Gender distribution analysis

All measures respond dynamically to the slicers placed on each report page, allowing users to filter results by Department, Gender, AgeGroup, PerformanceScore, and SalaryLevel. This interactivity is a core feature of the dashboard and is what enables exploratory analysis without requiring multiple static views.

6. Power BI Report Design

The report was originally designed as a single-page dashboard. After reviewing the volume of visualizations required, a decision was made to restructure it into a multi-page report to improve readability, reduce visual overload, and allow each analytical theme to be explored in greater depth.

The final report contains three pages, each focused on a distinct analytical area:

Page	Title	Focus Area
1	HR Analytics Overview	Workforce composition, KPIs, hiring trends, and distributions
2	Attrition Analysis	Employee turnover by salary, gender, department, and experience
3	Salary & Performance	Compensation patterns and salary-performance relationship

Design Principles

- Consistent color theme: A dark blue header on pages 2 and 3 with a clean white background for all chart areas.
- Navigation buttons: Page 1 includes two navigation buttons to access the Attrition and Salary & Performance pages. Pages 2 and 3 include a back button to return to the Overview.
- Slicers on every page: Five slicers (Department, Gender, AgeGroup, PerformanceScore, SalaryLevel) are present on all pages for consistent filtering.
- Minimal clutter: Each page limits the number of visuals to avoid information overload, grouping related charts in clearly defined sections.

7. Dashboard Page 1: HR Analytics Overview

The first page of the report serves as the main entry point and provides a high-level summary of the workforce. It is designed to answer the most immediate questions about the organization: how many employees, what they earn on average, how many have left, and how they are distributed across departments, genders, and age groups.

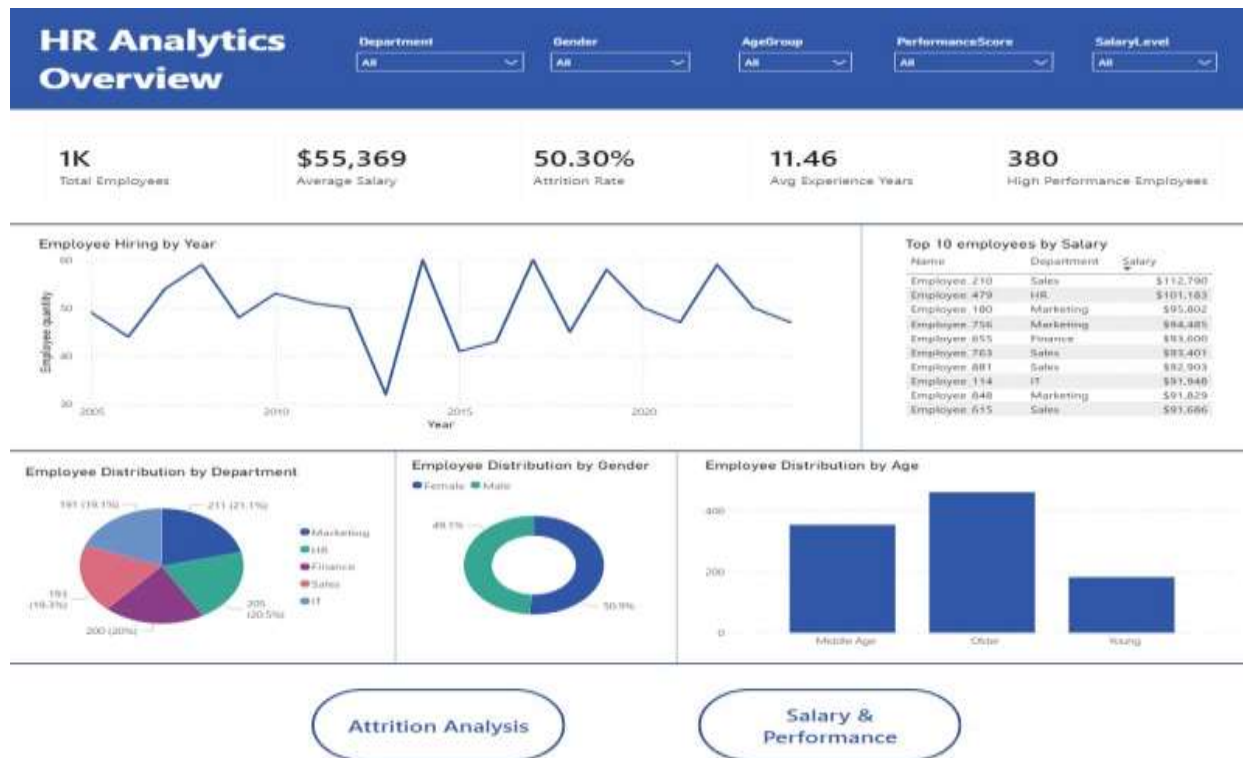


Figure 1: HR Analytics Overview Page

KPI Cards

Five KPI cards are displayed at the top of the page, providing an at-a-glance summary of the most important workforce metrics:

- Total Employees: 1,000
- Average Salary: \$55,369
- Attrition Rate: 50.30%
- Average Experience Years: 11.46
- High Performance Employees: 380

Employee Hiring by Year

A line chart showing the number of employees hired each year provides insight into the organization's historical growth and recruitment patterns. The chart covers hiring activity from approximately 2005 to 2023 and reveals fluctuations in annual hiring volume, including a notable dip around 2014-2015, which may indicate a period of reduced recruitment or organizational change. Without additional context, the cause of these variations cannot be confirmed.

Top 10 Employees by Salary

A table listing the ten highest-paid employees, including their name, department, and salary, is included on this page. The highest-paid employee (Employee 210, Sales) earns \$112,790, while the tenth-highest earns \$91,686. This table is useful for identifying high-compensation outliers and understanding where top salaries are distributed across departments.

Employee Distribution Visuals

- **Department:** A pie chart showing that Marketing is the largest department with 211 employees (21.1%), followed by HR (205, 20.5%), Sales (200, 20.0%), IT (193, 19.3%), and Finance (191, 19.1%). Departments are relatively evenly distributed.
- **Gender:** A donut chart showing a near-equal gender split of 50.9% Female and 49.1% Male.
- **Age Group:** A column chart showing that the Older (45+) group is the most common with 462 employees, followed by Middle Age (30-44) with approximately 357, and Young (18-29) with approximately 181.

Navigation

Two navigation buttons at the bottom of the page allow users to jump directly to the Attrition Analysis page or the Salary & Performance page without using the Power BI page tab navigation.

8. Dashboard Page 2: Attrition Analysis

The second page focuses entirely on employee attrition and is designed to help identify which workforce segments are at higher risk of turnover. Six visualizations are arranged across the page, examining attrition from multiple angles.

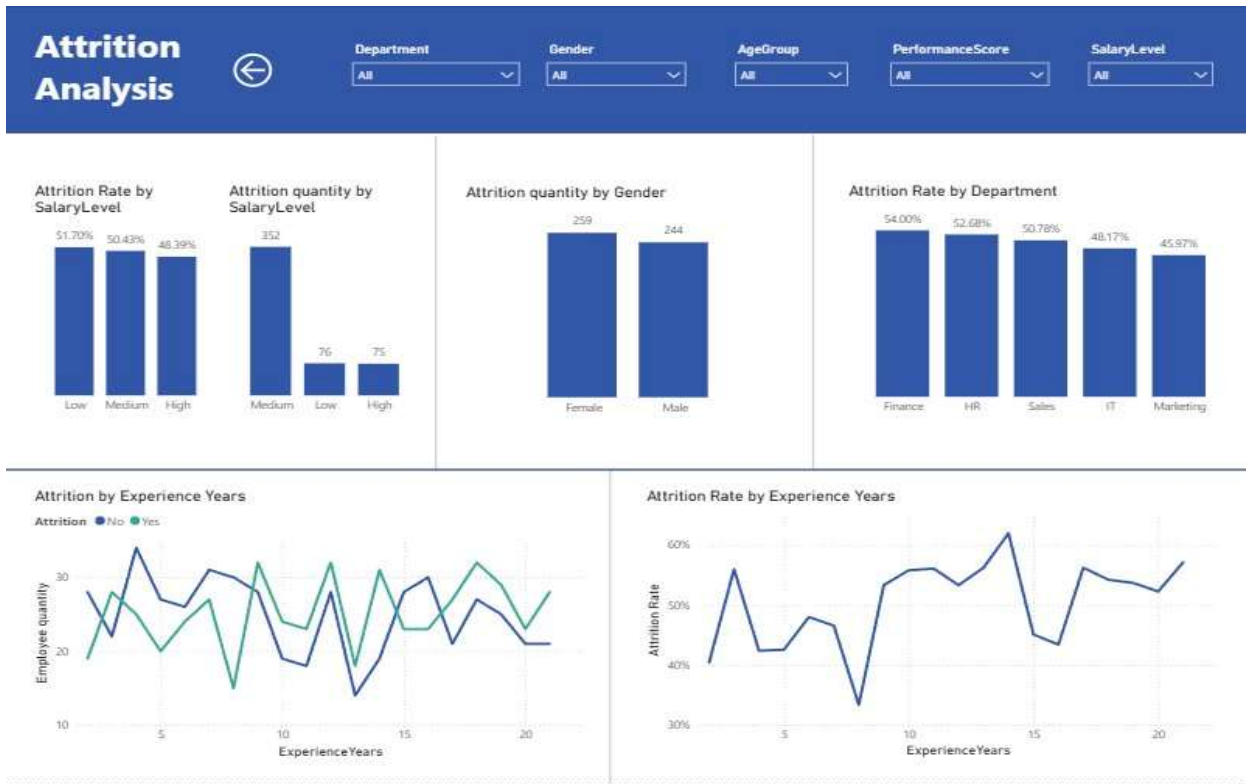


Figure 2: Attrition Analysis Page

Attrition by Salary Level

Two charts examine attrition across the three salary level groups. The attrition rate chart shows that the Low salary group has the highest attrition rate at 51.70%, followed by Medium at 50.43%, and High at 48.39%. This suggests that lower-compensated employees are marginally more likely to leave, though the difference between groups is relatively small and may not be statistically significant without further testing.

The attrition quantity chart reveals that the Medium salary group accounts for the largest absolute number of departures (352 employees), due to the greater number of employees in this band. The Low group accounts for 76 departures and the High group for 75.

Attrition by Gender

A bar chart shows that female employees account for 259 attrition cases compared to 244 for male employees. Given the near-equal gender split in the overall workforce, this difference is modest and does not appear to indicate a substantial gender-based attrition disparity.

Attrition Rate by Department

A bar chart ranking departments by attrition rate shows Finance as the highest at 54.00%, followed by HR at 52.68%, Sales at 50.78%, IT at 48.17%, and Marketing at 45.97%. Finance's elevated attrition rate, combined with its below-average salary, could suggest a link between compensation and retention in that department, though causal conclusions require additional investigation.

Attrition by Experience Years

Two charts at the bottom of the page examine attrition in relation to employee tenure. The line chart comparing attrition (Yes vs. No) by experience years shows interleaving patterns across the 0 to 20-year range, suggesting that attrition is not clearly concentrated at either end of the tenure spectrum. The attrition rate by experience chart fluctuates considerably, reaching peaks of approximately 55-60% at certain tenure points. This variability could reflect the synthetic nature of the dataset or may suggest that other factors, such as department or salary level, are driving attrition more than experience alone.

9. Dashboard Page 3: Salary & Performance

The third page focuses on compensation and performance patterns. It includes four visualizations examining salary by job title and department, the relationship between salary and performance score, and experience distribution by age group.

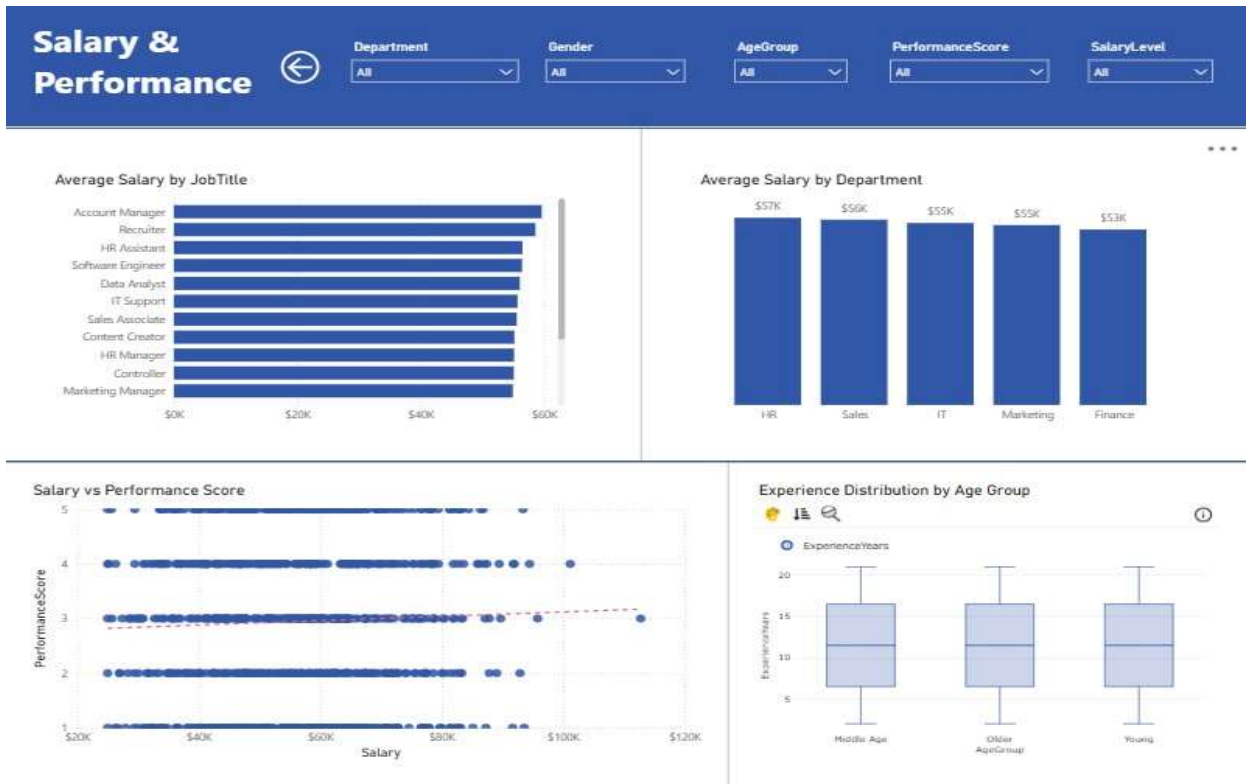


Figure 3: Salary & Performance Page

Average Salary by Job Title

A horizontal bar chart lists average salary by job title. Account Manager appears at the top of the list with the highest average salary, followed by Recruiter and HR Assistant. Marketing Manager and Controller appear at the lower end. The chart is useful for identifying which roles command higher compensation and may help in benchmarking against external salary data.

Average Salary by Department

A column chart shows average salary by department. HR leads with approximately \$57K, followed by Sales at \$56K, IT at \$55K, and Marketing at \$55K. Finance has the lowest average salary at approximately \$53K. This ranking is notable given that Finance also has the highest attrition rate, which could suggest that below-average compensation in that department may be contributing to turnover.

Salary vs Performance Score

A scatter plot visualizes the relationship between salary and performance score for all 1,000 employees, with a trend line added. The trend line is almost horizontal, indicating a weak positive relationship between the two variables. Employees with both high and low performance scores can be found at similar salary levels across the full salary range (\$20K-\$120K). This finding could suggest that compensation in this organization is not strongly linked to individual performance, which may indicate that tenure, job title, or department are more influential factors in determining salary. However, this conclusion requires further statistical analysis to confirm.

Experience Distribution by Age Group

A box plot shows the distribution of experience years across the three age groups (Middle Age, Older, Young). All three groups show median experience values of approximately 11-12 years, with similar interquartile ranges. This relatively uniform distribution is somewhat unexpected, particularly for the Young group, which would typically have lower tenure. This pattern may reflect the synthetic nature of the dataset.

10. Analytical Findings: Answers to 10 Key Questions

The following section provides direct answers to ten analytical questions based on the data and visualizations in the HR Analytics Dashboard.

Q1. Which department has the highest number of employees?

Marketing is the largest department, with 211 employees representing 21.1% of the total workforce. The other four departments (HR, Sales, IT, Finance) are relatively similar in size, ranging from 191 to 205 employees. The even distribution across departments could suggest that the organization maintains a balanced cross-functional structure.

Q2. Which department has the highest attrition rate?

Finance has the highest attrition rate at 54.00%, followed by HR at 52.68%. The lowest attrition rate is in Marketing at 45.97%. Based on the available data, Finance may warrant closer attention from an HR perspective. The combination of the lowest average salary and the highest attrition rate in Finance could suggest a compensation-retention link, though this requires further investigation before drawing firm conclusions.

Q3. Which gender is more represented in the company?

Female employees are slightly more represented, accounting for 50.9% of the workforce compared to 49.1% for male employees. The gender distribution is near-equal, which could reflect inclusive hiring practices. The attrition counts for both genders are also close (259 female, 244 male), suggesting no major disparity in retention between genders based on this dataset.

Q4. Which department has the highest average salary?

The HR department has the highest average salary, at approximately \$57,000. This is followed by Sales (\$56K) and IT (\$55K). Given that HR is also one of the mid-sized departments (205 employees), this finding may reflect the seniority of roles within the department or the job titles that are most common there.

Q5. Which department has the lowest average salary?

Finance has the lowest average salary, at approximately \$53,000. This is approximately \$4,000 below the HR department, which has the highest average. In combination with Finance having the highest attrition rate, this finding could suggest that salary levels in the Finance department may be a contributing factor to higher employee turnover, though causal relationships cannot be confirmed from this data alone.

Q6. Is there a relationship between PerformanceScore and Salary?

Based on the scatter plot on the Salary & Performance page, the relationship between PerformanceScore and Salary appears to be very weak. The trend line is almost horizontal, and salary values are distributed across a wide range at every performance score level. This could indicate that compensation in this organization is primarily driven by factors other than performance, such as job title, department, or tenure. Without additional statistical measures such as a correlation coefficient, it is not possible to confirm or quantify this relationship precisely.

Q7. Which age group is the most common?

The Older age group (45+) is the most common, with 462 employees representing approximately 46.2% of the workforce. The Middle Age group (30-44) accounts for approximately 35.7%, and the Young group (18-29) accounts for approximately 18.1%. The workforce appears to be skewed toward more experienced employees, which may have implications for succession planning and knowledge retention strategies in the future.

Q8. Who leaves the company more often — younger or more experienced employees?

Based on the attrition by experience years charts, there is no clear pattern suggesting that either very young employees or highly experienced employees leave at significantly higher rates. The attrition rate fluctuates across all experience levels and does not show a clear spike at either end of the spectrum. This could suggest that factors other than experience years, such as salary level, department, or role, are more influential drivers of attrition in this dataset. Further segmented analysis would be needed to confirm this.

Q9. How has the hiring trend changed over the years?

The Employee Hiring by Year line chart shows that hiring activity has been relatively consistent over the period covered by the dataset (approximately 2005 to 2023), with annual hiring counts generally ranging between 40 and 60 employees. A notable dip appears around 2014-2015, which could indicate a period of reduced recruitment, organizational restructuring, or budget constraints. Hiring levels appear to have recovered in subsequent years. Without additional context, the specific cause of this dip cannot be confirmed.

Q10. In which SalaryLevel group is attrition most common?

The Medium salary group accounts for the highest absolute number of attrition cases, with 352 employees having left. This is primarily because the Medium group contains the largest number of employees overall. However, in terms of attrition rate, the Low salary group has the highest rate at 51.70%, compared to 50.43% for Medium and 48.39% for

High. This may suggest that lower-compensated employees are marginally more likely to leave, though the differences between groups are small and should be interpreted cautiously.

11. Key Insights

The following key insights are drawn from the combined analysis across all three pages of the HR Analytics Dashboard:

- **The overall attrition rate of 50.30% is unusually high and may reflect the synthetic nature of the dataset. In a real-world context, this level of turnover would be a critical concern for any organization and would require immediate attention.**High overall attrition rate:
- **The Finance department shows the combination of the lowest average salary (~\$53K) and the highest attrition rate (54.00%), suggesting a potential compensation-retention challenge in this area that may benefit from further investigation.**Finance department risk:
- **The workforce shows a balanced gender split (50.9% female, 49.1% male), and attrition counts are similarly distributed, indicating no apparent gender-based retention disparity.**Near-equal gender distribution:
- **The scatter plot indicates that performance score is not strongly associated with salary level, with the trend line remaining nearly flat across all salary ranges. This could suggest that other factors drive compensation decisions in this organization.**Weak performance-salary link:
- **The dominance of the Older (45+) age group (46.2% of the workforce) may have long-term implications for succession planning and organizational knowledge retention.**Older workforce profile:
- **While the Low salary group has the highest attrition rate, the Medium group contributes the most departures in absolute numbers due to its larger size.**Medium salary group accounts for most attrition:
- **The spread in average salary across departments ranges from approximately \$53K (Finance) to \$57K (HR), a difference of roughly \$4K. This relatively narrow range could suggest a fairly uniform pay structure across departments, although job-title-level analysis reveals more variation.**Department salaries are relatively close:

12. Limitations

Several limitations should be considered when interpreting the findings of this project:

- **The dataset appears to have been generated artificially for educational purposes. Patterns such as the 50.30% attrition rate and the unusually uniform experience distribution across age groups may not reflect real-world HR dynamics.**Synthetic dataset:
- **The dashboard identifies associations and patterns in the data but does not establish causal relationships. For example, while Finance has both low average salary and high attrition, it cannot be confirmed from this data alone that compensation is causing the turnover.**No causal analysis:
- **The dataset does not include variables such as education level, marital status, job satisfaction, or employee location, which are commonly used in more comprehensive HR attrition models.**Limited demographic variables:
- **Attrition is recorded as a binary Yes/No variable without a date of departure. This limits the ability to analyze how attrition has evolved over time or to identify periods of elevated turnover.**No time-series attrition data:
- **It is not specified whether a score of 1 or 5 represents the highest performance. This analysis assumed a 1-5 scale where higher scores indicate better performance, but this assumption could not be confirmed from the dataset alone.**PerformanceScore scale ambiguity:
- **The report is built on a single flat table. A more robust data model using related tables (e.g., separate dimension tables for departments, job titles, and employees) could enable more complex and accurate analysis.**Single-table data model:

13. Recommendations

Based on the analytical findings from this project, the following recommendations are offered. These are intended as analytical suggestions based on the available data and should be validated against real-world context before any decisions are made.

- **Given that Finance has the lowest average salary and the highest attrition rate, a targeted compensation review for that department could be warranted. Even a modest salary adjustment may improve retention rates.**Review Finance department compensation:
- **The high overall attrition rate of 50.30% suggests that attrition is systemic and not limited to a single department or demographic group. A more detailed exit survey analysis or regression-based attrition model could help identify root causes.**Investigate attrition drivers more broadly:
- **With nearly half the workforce in the 45+ age group, the organization may face significant knowledge loss as these employees retire. Mentoring programs and structured knowledge transfer initiatives could help mitigate this risk.**Develop succession planning strategies:
- **The weak relationship between PerformanceScore and Salary could suggest that performance-based pay structures are not fully implemented or effective. Reviewing the compensation policy to ensure that high performers are adequately rewarded may help improve both retention and motivation.**Reconsider performance-linked compensation:
- **Incorporating additional variables such as exit interview data, job satisfaction scores, and department-level budget information would significantly enrich the analytical depth of future HR dashboards.**Enhance the data model for future analyses:

14. Conclusion

This project demonstrates the end-to-end development of an HR analytics solution using Microsoft Power BI. Starting from a raw CSV dataset containing 1,000 employee records, the project involved data cleaning and transformation in Power Query, DAX measure creation, and the design of a three-page interactive dashboard covering workforce overview, attrition analysis, and salary and performance.

The analytical findings highlight several notable patterns in the data, including a high overall attrition rate, a Finance department that shows both below-average compensation and above-average turnover, a workforce skewed toward older employees, and a weak relationship between performance scores and salary levels. While these findings are based on a synthetic dataset and should not be interpreted as representative of a real organization, they provide a meaningful context for practicing and demonstrating HR analytics skills.

The project successfully demonstrates proficiency in Power BI, Power Query, DAX, and dashboard design, and the resulting report is suitable for both academic submission and professional portfolio presentation.